UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Convert 286610 to octal. Show your complete workflow.

QUESTION A2 **(5 marks)**

The Difference Engine was a mechanical computer designed by Charles Babbage in the 19th century. It was one of the earliest examples of a mechanical calculator, built to automatically compute mathematical tables for naval navigation.

a) What device was the successor of the Difference Engine?

b) Describe the difference in function between the Difference Engine and Mechanical Calculating Machine.

c) Briefly explain the available technology that existed in the computer generation of Difference Engine.

QUESTION A3 **(5 marks)**

a) List THREE (3) characteristics of Symmetric Multiprocessor (SMP).

b) Describe ONE (1) advantage of Symmetric Multiprocessor (SMP) interms of availability.

QUESTION A4 **(5 marks)**

Given the following Figure 2.

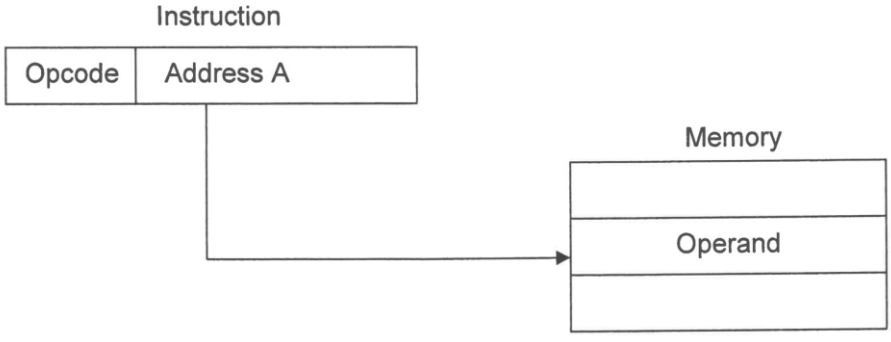


Figure 2

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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

QUESTION A5 **(10 marks)**

a) Discuss THREE (3) main key factors for memory selection.

b) List TWO (2) examples for internal cabling memory and TWO (2) examples of system memory.

QUESTION A6 **(10 marks)**

a) Differentiate the involvement of CPU in the interrupt-driven I/O technique and Direct Manipulation Access

b) Define interrupt.

c) Discuss what happens during a context switch.

d) Elaborate on the benefit of a context switch.

QUESTION A7 **(10 marks)**

a) Draw a complete machine cycle by giving the exact location of each phase.

b) Briefly, describe each phase in the machine cycle.

QUESTION A8 **(10 marks)**

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

QUESTION A9 **(10 marks)**

a) Describe THREE (3) types of interfaces to external devices.

b) Explain TWO (2) types of interrupt.

QUESTION A10 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 156510 to hexadecimal.

b) Convert BDC16 to decimal.

a) Convert 156510 to hexadecimal.

b) Convert BDC16 to decimal.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 1 and Table 1.

$$z = A^2 - B * C + \frac{2}{D + 5}$$

Figure 1

write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three- address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.

Table 1: Machine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	suB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	